

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Y. Greeshma Reddy	Reg. No.:	192424146
Section / Batch:	B. Tech-AI&DS	Date:	16-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 – 50 Q2 – 50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Cybersecurity Threat Analysis System A cybersecurity system analyzes threats using the recurrence $T(n)=T(n-1)+n\log n$. Apply the substitution method to solve the recurrence. Expand step-by-step and derive the complexity. Analyze how the algorithm scales with increasing threat data. Interpret efficiency using asymptotic notation.

Code of Conduct

I Y. Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Greeshma Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Cybersecurity Threat Analysis System

Problem:

A cybersecurity system analyzes threats using the recurrence:

$$T(n) = T(n - 1) + n \log n$$

Apply the substitution method, derive the complexity, analyze scalability, and interpret using asymptotic notation.

1. Understanding of Problem Context

- The cybersecurity system processes threat records sequentially.
- Each new threat requires processing the previous results plus additional analysis.
- The recurrence represents cumulative processing over all threat data.
- As threat volume increases, computation also increases.
- The objective is to determine the algorithm's overall time complexity.

2. Application of Concepts

- Applied the **Substitution Method** to expand the recurrence.
- Repeatedly substituted until reaching the base case.
- Converted the recurrence into a summation.
- Used summation properties to simplify the expression.
- Applied asymptotic notation to express the final complexity.

3. Problem-Solving Approach

- Write the recurrence:

$$T(n) = T(n - 1) + n \log n$$

- Expand recursively:

$$T(n) = T(n - 2) + (n - 1)\log(n - 1) + n \log n$$

- Continue expansion:

$$T(n) = T(1) + \sum_{i=2}^n i \log i$$

- Approximate the summation:

$$\sum i \log i = \Theta(n^2 \log n)$$

- Therefore,

$$T(n) = \Theta(n^2 \log n)$$

4. Accuracy of Solution

- The recurrence was expanded correctly.
- The summation form was obtained accurately.
- Growth of the summation was evaluated correctly.
- Final time complexity is:

$$\boxed{\Theta(n^2 \log n)}$$

- The solution satisfies the substitution method.

5. Clarity

- Each substitution step was shown clearly.
- Mathematical derivation followed a logical sequence.
- Complexity was expressed using asymptotic notation.
- Scalability analysis was explained in simple terms.
- The conclusion clearly states that execution time increases rapidly for large threat datasets.